

CMSC 10 – INTRODUCTION TO COMPUTER SCIENCE

Course Credit: 3 units

Course Description: Introduction to the major areas of computer science – software systems and methodology, computer theory, computer organization and architecture – and relevant social issues.

Objectives: By the end of the semester, students would have been introduced to the basic concepts of computer science. Specifically, the students would be able to:

1. Define computer science;
2. Design algorithms for a variety of problems;
3. Describe the structure and organization of computers;
4. Explain the importance of systems software in the execution of programs;
5. Differentiate the major computer applications; and
6. Discuss relevant social issues.

Course Prerequisite: None

Instructional Format: The course will involve lecture sessions to introduce the different concepts. There will be quizzes, lecture discussions, and exams to verify students' understanding of the corresponding covered topics. Students are encouraged to ask questions during the classes and during the consultation hours of the faculty-in-charge.

Attendance: As per university rule, a student must not miss more than 20% of the total number of class sessions (7 sessions). Going beyond the maximum number of allowable absences, in either lecture or laboratory, means a grade of either DRP (if most are excused) or 5.0 (if most are unexcused). Perfect attendance does not merit extra points in the grade. It is important that students submit their excuse slip/letter (if applicable) as soon as possible after being absent. Failure to do this is a sign of irresponsibility.

Grade of 4.0 and INC: There will be no grades of 4.0 and INC. A student who completes the requirements of the course will be given a passing mark. Failure to comply with the requirements of the course within the semester will mean a grade of 5.0.

Dropping: A student is officially dropped from the course if he/she successfully completes the dropping process. A student who stops attending classes and/or taking exams without officially dropping will be given a grade of 5.0. Note that the teacher has the right not to sign your dropping slip.

Missed Quizzes and Exams: A student who missed an exam should immediately inform the faculty-in-charge of the course. Students will only be able to take the missed exam if their absence is deemed excusable as approved by the Office of the College Secretary (OCS). Schedule of the examination will be set on a later date upon presentation of the OCS-approved excuse slip. There will be no make-up quizzes.

Academic Dishonesty: All forms of academic dishonesty will not be tolerated. Students caught cheating will automatically get a grade of 5.0 for the course, and the teacher may file disciplinary charges against the student(s).

Schedule of Classes: 2:30-4:00 TTh.

Grading Scheme:

Quizzes/Assignments -	50%
AI Project -	50%
Long Exam 1 -	100%
Long Exam 2 -	100%
Average -	100%

Grading Scale:

From	To	Grade
96	100	1.00
92	95.99	1.25
88	91.99	1.50
84	87.99	1.75
80	83.99	2.00
75	79.99	2.25
70	74.99	2.50
65	69.99	2.75
60	64.99	3.00
55	59.99	4.00
0	54.99	5.00

Lecture Outline:

1. Introduction (Nature of Computer Science)
 - a. What is Computer Science?
 - b. History and Evolution of Computers
 - c. Algorithms and Problem Solving
2. Software/Hardware Hierarchy
 - a. Evolution and Types of computers
 - b. Relationship between hardware and software
 - c. Structure/organization and components of computers
 - d. Capabilities and Limitations of Computers
 - e. Categories of software
 - f. Operating System
 - g. Programming Languages
 - h. Language Implementation Methods
 - i. Application Programs

3. Machine Organization
 - a. Decimal, Binary, Octal and Hexadecimal number Systems
 - b. Operations of Number Systems
 - c. Conversion from one number system to another
 - d. Digital Circuits & Boolean Algebra
 - AND Gate
 - OR Gate
 - NOT Gate
4. Elements of Programming
 - Design of Algorithm
 - Writing Algorithms
 - Implementation of Algorithms
 - Techniques in Program Development
 - Modular Programming
 - Top-Down Design
 - Bottom-Up Design
 - Functions (Passing Value and Reference)
 - Parameters (Variables / Identifiers Scope)
4. Elements of Programming
 - a. Design of Algorithm
 - b. Writing Algorithms
 - c. Implementation of Algorithms
 - d. Techniques in Program Development
5. Introduction to Database Management System
 - a. Understanding Data and Information
 - b. Data types & representation
 - c. Introduction to databases and DBMS
 - d. Relational database basics
 - e. Data cleaning and data organization
 - f. Basic Data Retrieval & Analysis
6. Theory of Algorithms
 - a. Characteristics of Algorithms
 - b. Algorithm Design Techniques
 - c. Complexity of Algorithms
 - d. Computability
7. Introduction to Artificial Intelligence
 - a. AI Fundamentals
 - b. How AI Learn
 - c. Generative AI and LLMs
 - d. AI Applications and Building with AI
 - e. Future of AI

8. Impacts of Computers to the Society
 - a. Computers and Society
 - b. Computer Crime and Security
 - c. Privacy
 - d. Intellectual Property
 - e. Responsible AI

References:

- [1] Brookshear, G. and Brylow, D. (2014). *Computer Science: An Overview*, 12th Ed. Pearson.
- [2] Mata-Toledo, R. and Cushman, P. K. (1999). *Introduction to Computer Science*. McGraw-Hill.
- [3] Cormen, T.H., Leiserson, C. E., Rivest, R. L., and Stein, C. (2009). *Introduction to Algorithms*, 3rd Ed. MIT Press and McGraw-Hill.
- [4] Mano, M. M. and Ciletti, M. D. (2006). *Digital Design*, 4th Ed. Prentice Hall.
- [5] Abd-El-Barr, M. and El-Rewini, H. (2004). *Fundamentals of Computer Organization and Architecture*. Wiley-Interscience.
- [6] Sebesta, R. (2015). *Concepts of Programming Languages*, 11th Ed. Pearson.
- [7] Stallings, W. and Brown, L. (2017). *Computer Security: Principles and Practice*, 4th Ed. Pearson.
- [8] Reynolds, G. W. (2018). *Ethics in Information Technology*, 6th Ed. Course Technology.

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Session (Day #)	Course Outcomes	Topics	Activities	Reference Materials	Assessment
1	Identify the course requirements and explain the scope, expectations, and assessment scheme of CMSC 10.	Course Orientation and Syllabus	Course orientation; expectations setting	Course Syllabus	Class participation
2-3	Define computer science and explain its major areas, historical development, and role in problem solving.	Introduction to Computer Science: nature of CS; history and evolution of computers; algorithms and problem solving	Interactive lecture; concept mapping; short problem-solving activity	[1], [2]	Class participation; short exercise
4-5	Explain the relationship among hardware, software, operating systems, programming languages, and application programs.	Software/Hardware Hierarchy: computer types and components; hardware-software relationship; software categories; operating systems; programming languages; applications	Lecture-discussion; computer-system identification activity	[1], [2], [5], [6]	Class participation; Quiz 1
6-8	Represent information using common number systems and explain the role of digital logic in computer systems.	Machine Organization: decimal, binary, octal, and hexadecimal systems; conversions and operations; Boolean algebra; AND, OR, and NOT gates	Lecture-demonstration; board work; guided exercises	[1], [2], [4], [5]	Problem set; Quiz 2
9-10	Design and express algorithms for simple computational problems.	Elements of Programming I: algorithm design; pseudocode/flowchart; writing algorithms	Lecture-discussion; board work; individual algorithm exercises	[1], [2]	Class participation; programming exercise

Session (Day #)	Course Outcomes	Topics	Activities	Reference Materials	Assessment
11-12	Explain how algorithms are translated into programs and apply basic program-development approaches.	Elements of Programming II: implementation of algorithms; program-development techniques; modular, top-down, and bottom-up design	Lecture-discussion; guided design activity	[1], [2], [6]	Quiz 3; design exercise
13-15	Explain functions, parameters, scope, and basic notions of program correctness.	Elements of Programming III: functions; parameters; variable/identifier scope; program correctness	Lecture-demonstration; tracing exercises; problem solving	[1], [2], [6]	Class participation; Quiz 4
16	Demonstrate understanding of the concepts covered in Sessions 1-15.	LONG EXAM 1	Written examination		Long Exam 1
17-18	Distinguish data from information; explain basic DBMS concepts; organize, clean, retrieve, and summarize simple datasets.	Introduction to Data and Database Management Systems: data and information; data types and representation; databases and DBMS; relational database basics; data cleaning and organization; basic data retrieval and analysis	Lecture-demonstration; hands-on data cleaning; simple table design; basic query/filtering activity	[1], [2]	Hands-on activity; Quiz 5
19-20	Describe the characteristics of algorithms and explain basic ideas of algorithm design, complexity, and computability.	Theory of Algorithms: characteristics; design techniques; complexity; computability	Lecture-discussion; algorithm comparison exercises	[1], [2], [3]	Class participation; Quiz 6

Session (Day #)	Course Outcomes	Topics	Activities	Reference Materials	Assessment
21	Explain foundational concepts of artificial intelligence and distinguish AI from conventional computing approaches.	Introduction to Artificial Intelligence I: AI fundamentals; major approaches and applications	Lecture-discussion; AI application identification activity	[1], [2]	Class participation
22	Explain, at an introductory level, how AI systems learn from data and recognize the role of training, models, and inference.	Introduction to Artificial Intelligence II: how AI learns; data, training, models, and inference	Lecture-demonstration; guided examples	[1], [2]	Short exercise; Quiz 7
23	Explain generative AI and large language models, including their capabilities and limitations.	Introduction to Artificial Intelligence III: Generative AI and Large Language Models	Lecture-demonstration; prompt activity; output evaluation	[1], [2]	Class participation; AI activity
24	Apply introductory AI concepts in defining a simple AI-enabled solution to a real-world problem.	Introduction to Artificial Intelligence IV: AI applications; building with AI; future of AI	Project workshop; problem definition; solution design	[1], [2]	AI Project proposal/checkpoint
25-26	Discuss the social impacts of computing and identify major concerns involving cybersecurity and privacy.	Impacts of Computers on Society I: computers and society; computer crime and security; privacy	Case discussion; scenario analysis	[7], [8]	Case analysis; Quiz 8

Session (Day #)	Course Outcomes	Topics	Activities	Reference Materials	Assessment
27	Explain intellectual property issues relevant to computing and digital content.	Impacts of Computers on Society II: intellectual property	Lecture-discussion; case analysis	[7], [8]	Class participation
28	Evaluate ethical and responsible-use issues arising from artificial intelligence and computing technologies.	Impacts of Computers on Society III: Responsible AI; fairness, transparency, accountability, privacy, and human oversight	Structured discussion; AI ethics case study	[7], [8]	Reflection/case response
29-31	Integrate computing, data, and AI concepts in developing a simple solution or prototype.	AI Project Development and Integration	Project workshop; consultation; prototype development and testing	Course materials	AI Project progress
32	Communicate and demonstrate an AI-enabled solution and explain its computing concepts, data use, limitations, and social implications.	AI Project Presentation and Demonstration	Project presentation; peer feedback	Course materials	AI Project
33	Synthesize and review the major concepts of computer science covered in the course.	Course Integration and Review	Integrative review; concept mapping; question-and-answer session	[1]-[8]	Review exercise
34	Demonstrate understanding and integration of the major concepts covered after Long Exam 1.	LONG EXAM 2	Written examination		Long Exam 2

Revision History

2025 Sep 2

2026 August 27 – include introduction to DBMS

2026 September 1 – enhanced weekly schedule; integrated DBMS, AI, project activities, and two-exam structure